

Instagram Account Mirror

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Mission: local mirror of one Inst. account but with gateway for comments

Scope:

- full sync of all user media from Inst. Graph API (POST /api/sync/)
- paginated read to all saved local posts (GET /api/posts/)
- posting a comment to Inst. after saving locally (POST /api/posts/{id}/comment/)
- Docker compose for app + Postgre
- Integration tests

Not in Scope:

- OAuth + token mgmt
- auto sync (only manual pull)
- media storage

Related Projects:

- Inst. Graph API
- Django REST
- DM service

System Context:

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Client (-s)

HTTP

App

Views

Services
(bus. logic)

Inst. Client
(HTTP OUT)

DB

post | comment

Inst. Graph API

Design:

Class Post:

id: int

instagram_id: str

caption: str | None

media_type: Enum { IMAGE, VIDEO, CAROUSEL_ALBUM }

media_url: str | None

permalink: str

timestamp: datetime

synced_at: datetime

@invariance:

instagram_id: $|\{p.id \mid p \in \text{Posts}\}| = |\mathcal{P}|$


```

timestamp <= synced_at
func from_inst_payload(data: dict) → Post:
    @precond:
        id, permalink, timestamp, media_type ∈ data
    @postcond:
        return Post with self.inst_id = data['id']
func upsert(data: dict) → Post, created: bool
    @precond:
        id, permalink, timestamp, media_type ∈ data
    @postcond:
        if ∃ p: Post s.t. p.inst_id = data['id']
            then p is updated, created = False
        else : new Post, created = True
    synced_at = now()

```

Class Comment:

```

id: int
post: Post
inst_comment_id: str
text: str
created_at: datetime
@invariance:
    len(text) ≥ 1 and len(text) ≤ 2200
    inst_comment_id != ""

```


post $\exists$ in DB

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func create(post: Post, text: str, inst_comment_id: str) $\rightarrow$ Comment:

@precond:

post.pk is not None

inst_comment_id $\neq$ ""

$1 \leq \text{len}(\text{text}) \leq 2200$

@postcond:

self.post = post

self.text = text

self.pk is not None

Class InstagramClient:

-- init -- (access_token: str, api_version: str = "v21.0"):

@precond:

access_token $\neq$ ""

@postcond:

self._token = access_token

func fetch_all_media(fields: list[str]) $\rightarrow$ list[dict]:

@precond:

self._token is valid

@postcond:

returns paginated list of all media dict-s
 $\text{len}(\text{result}) = \text{total media objects}$

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raises Instagram API Error if request fails
func publish_comment(media_id: str, text: str) → str:
@precond:

media_id != ""

1 <= len(text) <= 2200

media ∉ on Inst

comments enabled

@postcond:

comment posted on Inst.

returns inst_comment_id

raises Instagram API Error on 400/500 from API

Class SyncService:

-- init -- (client: InstagramClient):

@precond:

client is valid

func sync_all() → SyncResult:

@post:

* m (media) returned by client. fetch_all_media
Post.upsert(m) was called

result.updated + result.created = total media retr

Inst API Error propagated from client

class SyncResult:

created: int

updated: int

Class Comment Service:

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--init-- (client: Just Client):

func post-comment (post-pk: int, text: str) →

Comment:

@precond:

Post.objects(pk=post-pk).exists() (or Post Not Found)

$1 \leq \text{len}(\text{text}) \leq 2200$

@postcond:

comment created on Just;

comment.post.pk = post-pk

* returns Comment instance

* raises Post Not Found if no Post with post-pk
or Just API Error if Graph API rejects the comment

UML Sequence Diagram:

UML Sequence Diagram:

1: POST /sync/

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Client SyncView SyncService DB Inst Client Graph API



